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TASK 6: DATABASE DESIGN AND IMPLEMENTATION

Presented by

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1. INTRODUCTION

This report presents a comprehensive design for an attendance management system that uses fingerprint authentication to track student attendance. The design process includes various stages: requirement analysis, conceptual design, logical design, physical design considerations, and normalization. Each stage is discussed in detail to ensure clarity and completeness. Key terms related to database design are also defined to provide a clear understanding of the concepts used in this report.

1.1. Definition of Key Terms

- **Database:** A database is a structured collection of data that is stored and accessed electronically. It allows for efficient data management, retrieval, and manipulation.
- **Entity:** An entity is an object or concept about which data is stored in a database. For example, in a university database, entities could include students, courses, and lecturers.
- **Attribute:** An attribute is a property or characteristic of an entity. For instance, a student entity may have attributes such as student ID, name, and email.
- **Primary Key (PK):** A primary key is a unique identifier for a record in a database table. It ensures that each record can be uniquely identified.
- **Foreign Key (FK):** A foreign key is an attribute that creates a relationship between two tables. It references the primary key of another table to establish a link between them.
- **ER Diagram (Entity-Relationship Diagram):** An ER diagram is a visual representation of the entities in a database and their relationships.

2. STAGES IN DATABASE DESIGN

1) Requirement Analysis
2) Conceptual Design
3) Logical Design
4) Physical Design

2.1. Requirement Analysis

In this phase, we understand the system requirements and identify key entities and relationships. The main requirements for the attendance management system include:

- ✓ **User Management:** Managing different types of users (students, lecturers, administrators).
- ✓ **Attendance Tracking:** Tracking student attendance in various courses.
- ✓ **Fingerprint Authentication:** Using fingerprint data to authenticate users.
- ✓ **Course Management:** Managing courses and their respective departments.
- ✓ **Administrative Functions:** Enabling administrators to manage users and generate reports.

2.1.1. Database Requirements

- **Security:** The database must prioritize data security, especially regarding sensitive student fingerprint information. Encryption techniques should be implemented to protect this data.
- **Performance:** The database should handle efficient fingerprint matching and attendance record retrieval for a large number of students.
- **Scalability:** The design should accommodate future growth in student population and potential integration of additional features.

Key entities identified from these requirements are:

- User
- FingerprintData
- Admin
- Student
- Lecturer
- Department

- Course
- AttendanceRecord

2.1.2. Database Model

The proposed database utilizes a relational model with 8 tables:

2.2. Conceptual Design

The conceptual design phase involves creating a high-level data model, often represented as an ER diagram. From the class diagram we can get a general view of how our design will look like since most of the entities and attributes are already found in the class diagram. The ER diagram for the attendance management system is as follows:

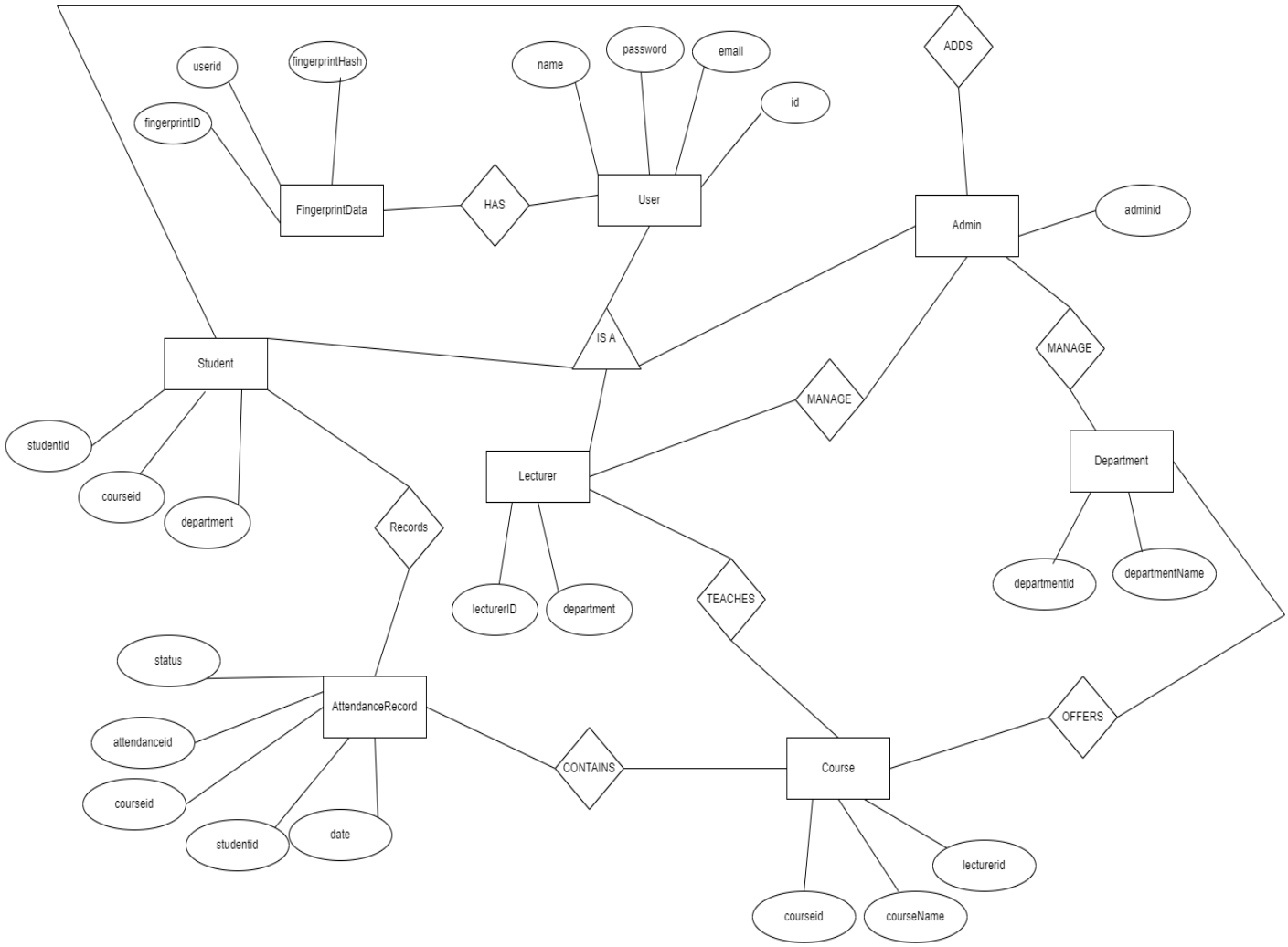


FIG 1: ER DIAGRAM

2.3. Logical Design

The logical design phase involves translating the conceptual model into a relational schema. The schema is structured to ensure data integrity and efficient querying. We have the following features in our relational schema;

- **Tables:** Each entity becomes a table (User, FingerprintData, Department, etc.).
- **Primary Keys (PK):** Each table has a unique identifier (e.g., studentId for Student).
- **Foreign Keys (FK):** Relationships are established using foreign keys. For example, AttendanceRecord references studentId (Student) and courseId (Course).

2.3.1. Normalization

Normalization ensures the schema adheres to principles that avoid redundancy and maintain data integrity. The schema is designed to be in the third normal form (3NF):

- **First Normal Form (1NF):** All tables contain atomic (indivisible) values.
- **Second Normal Form (2NF):** All non-key attributes are fully functionally dependent on the primary key.
- **Third Normal Form (3NF):** All attributes are functionally dependent only on the primary key, with no transitive dependencies.

Below is a diagram representing the relational schema:

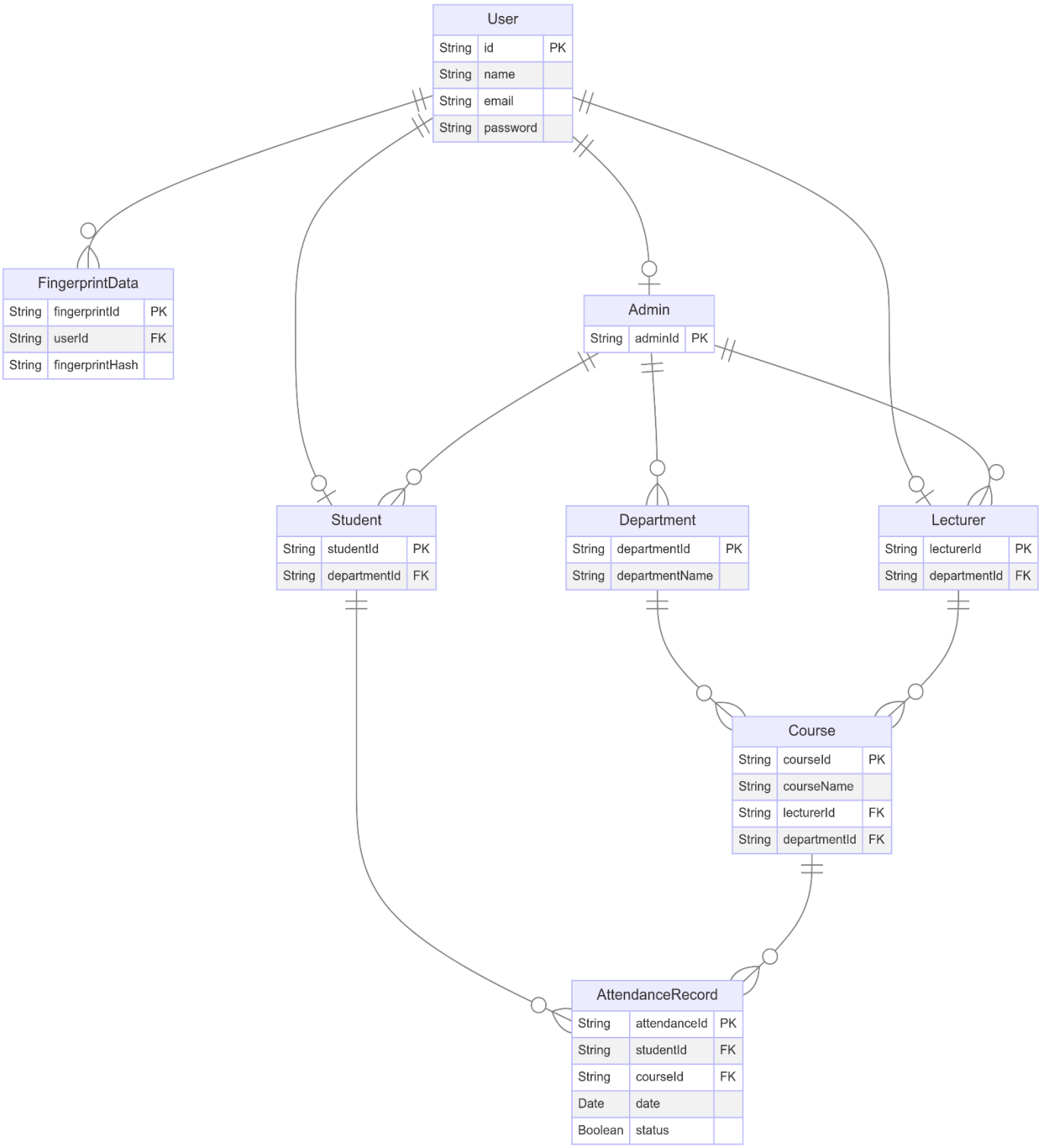


FIG 2: RELATIONAL SCHEMA

2.4. Physical Design

The logical schema is further defined for a specific database management system (DBMS) like MySQL or PostgreSQL:

- **Data Types:** Columns are assigned specific data types based on the data they store (e.g., VARCHAR for names, DATE for attendance date).
- **Constraints:** Additional constraints can be applied (e.g., CHECK constraints to ensure boolean values in AttendanceRecord.status).
- **Indexes:** Indexes can be created on frequently used columns to improve query performance (e.g., studentId in AttendanceRecord).
- **Data Security:** The following key points are noted during security analysis

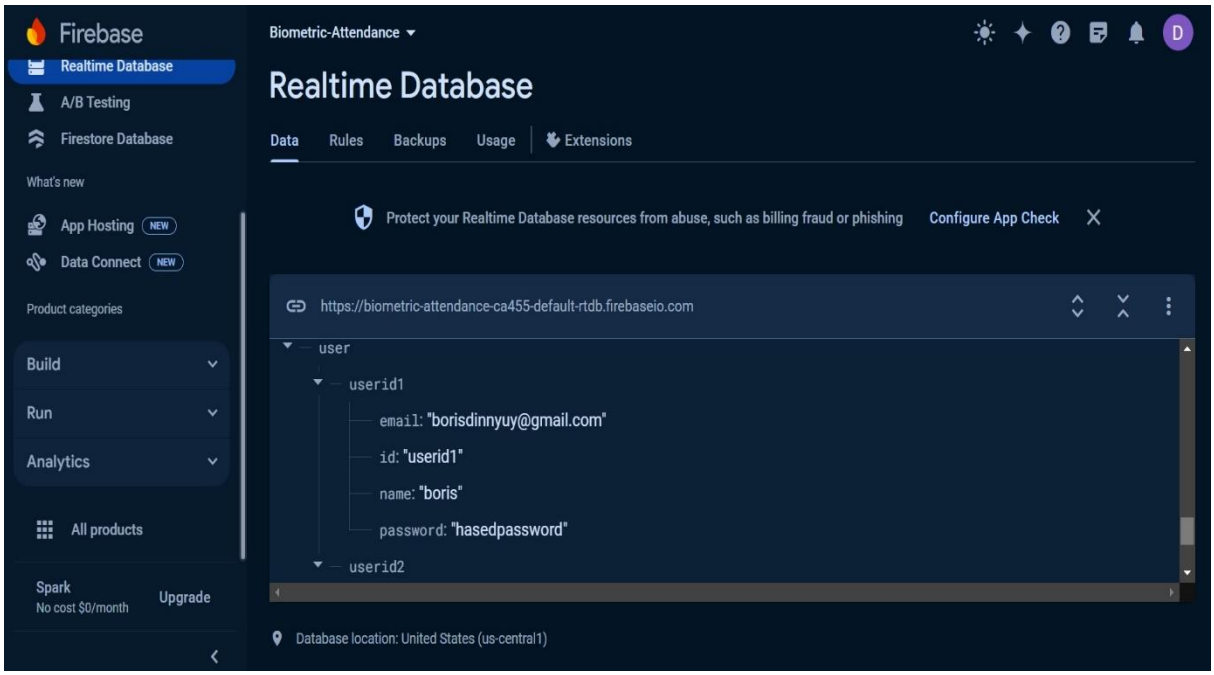
    1. **Fingerprint Storage:** Fingerprint data should not be stored directly in the database. Instead, a one-way hashing function should be used to create a unique identifier (template) from the fingerprint data. This template cannot be reversed to recreate the original fingerprint data.
    2. **Encryption:** Sensitive data fields, such as password and contact information, should be encrypted at rest and in transit.
    3. **Access Control:** Implement a user access control system with different permission levels. Grant access to attendance data only to authorized personnel (lecturers, administrators).

3. IMPLEMENTATION

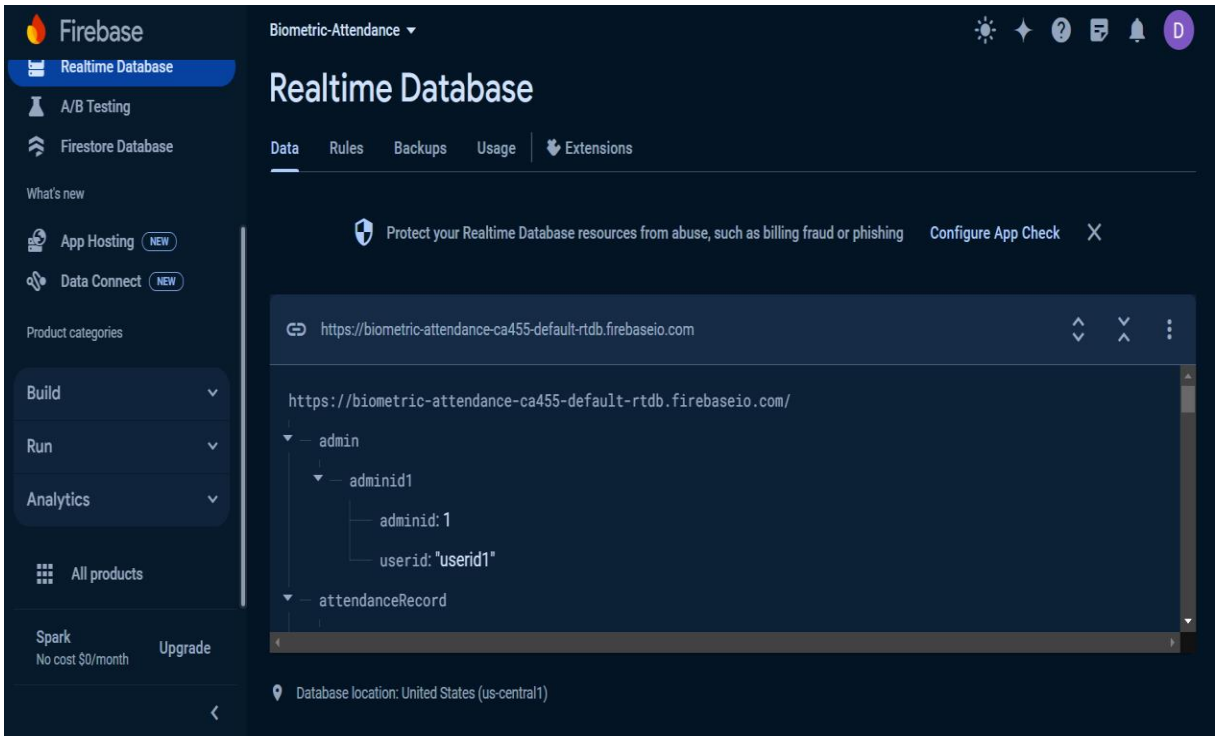
We used **Firebase** for our database implementation after passing through a thorough design and ensuring that our database delivers user expectations. Some of the main functionality it provides to developers includes a real-time database, file storage, user authentication and crash reporting tools. The Firebase Realtime Database is a cloud-hosted NoSQL database that enables easy real-time syncing of data between clients.

Below are the different collections created

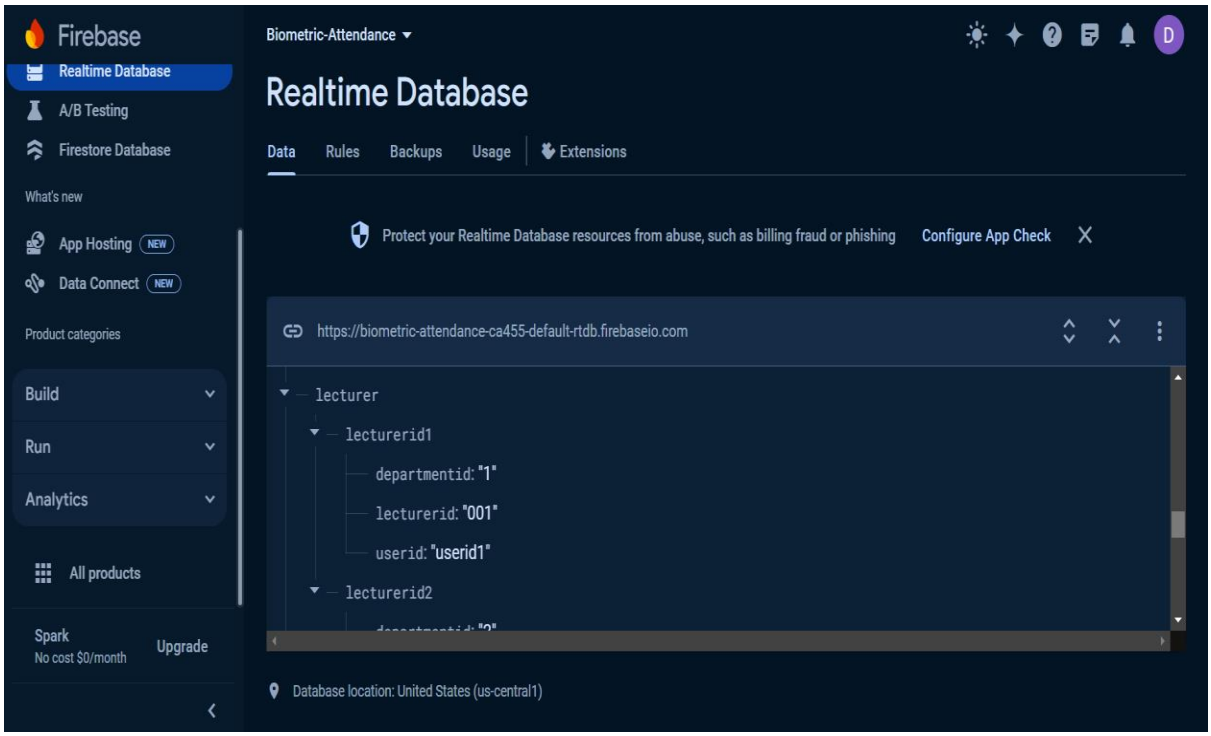
1. User collection



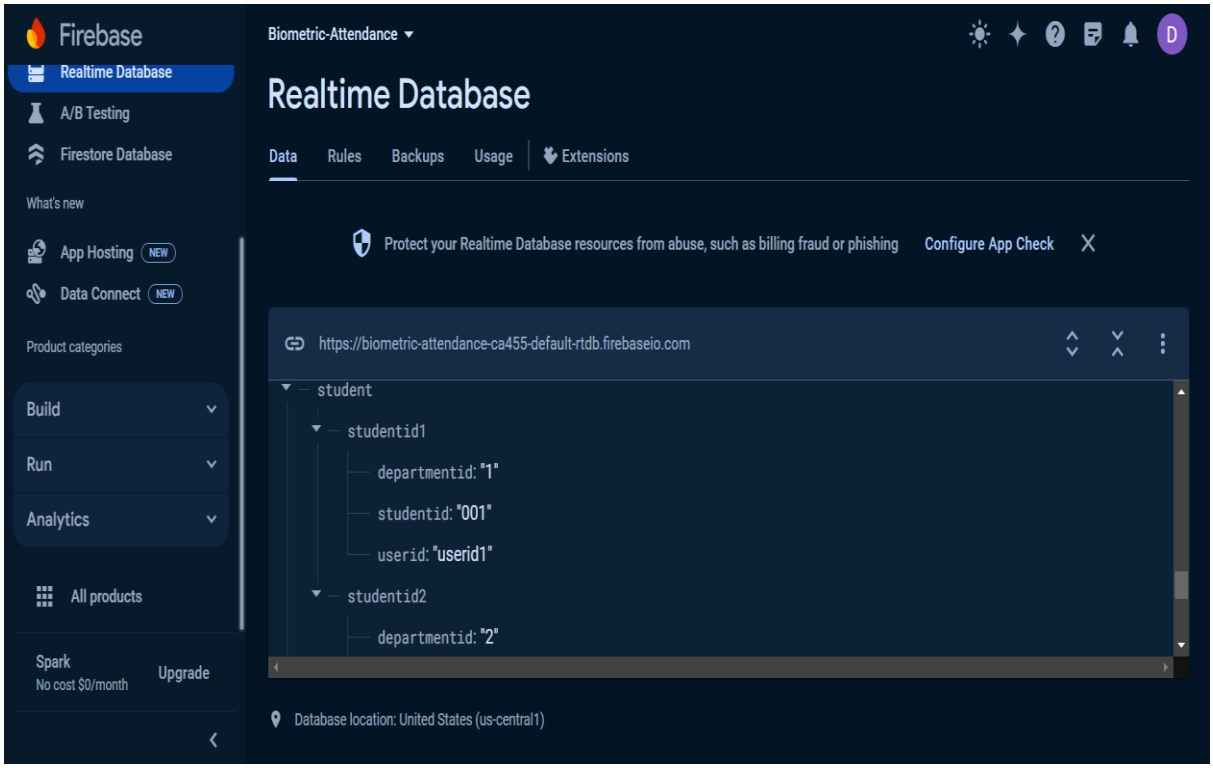
2. Admin collection



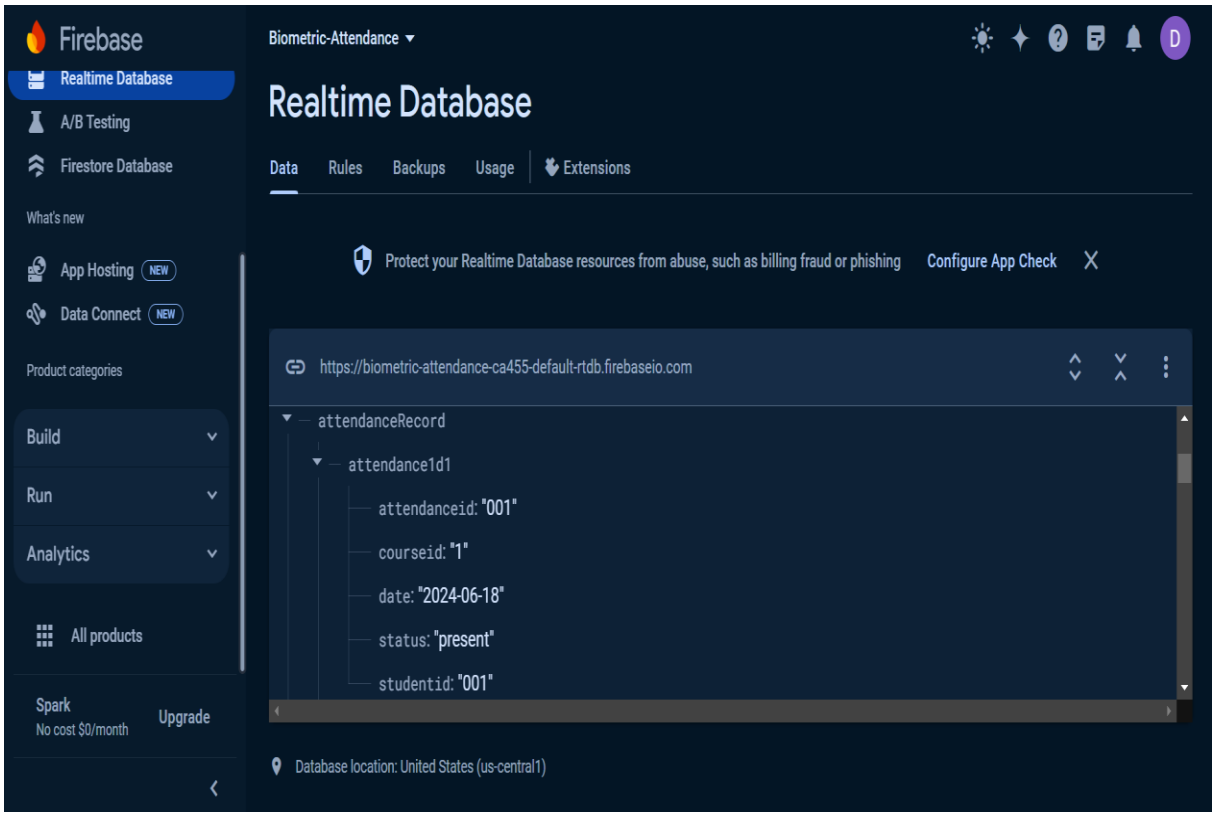
3. Lecturer collection



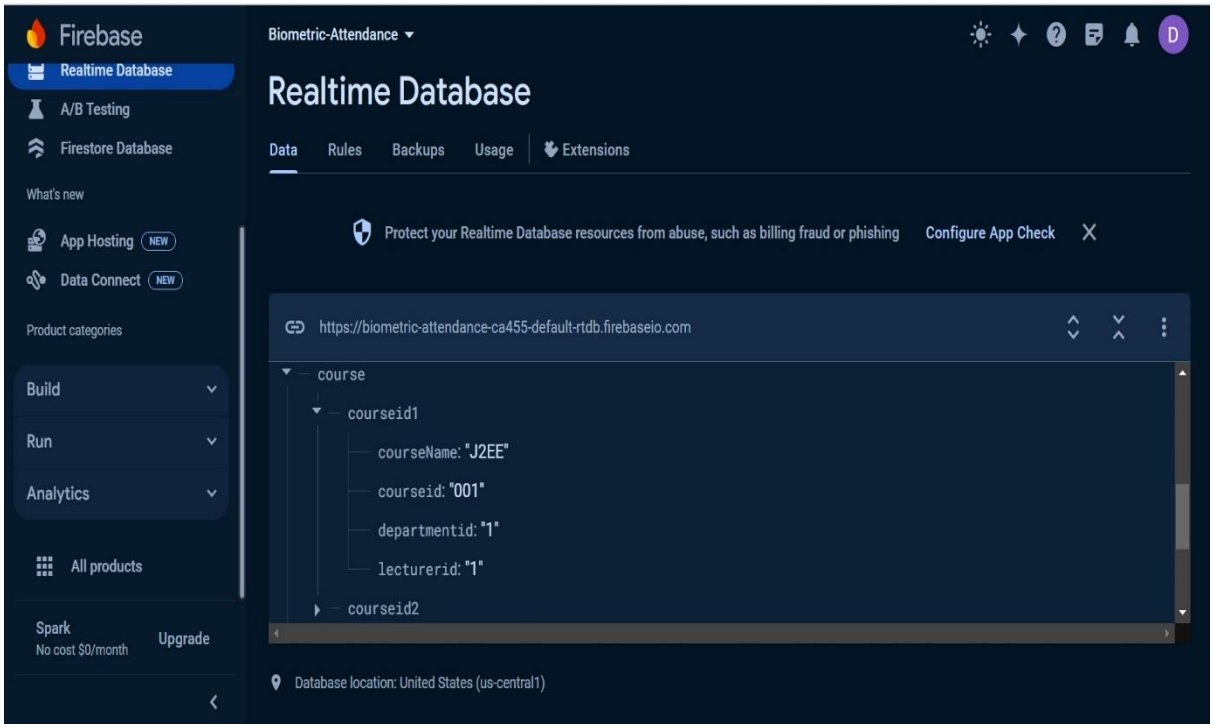
4. Student collection



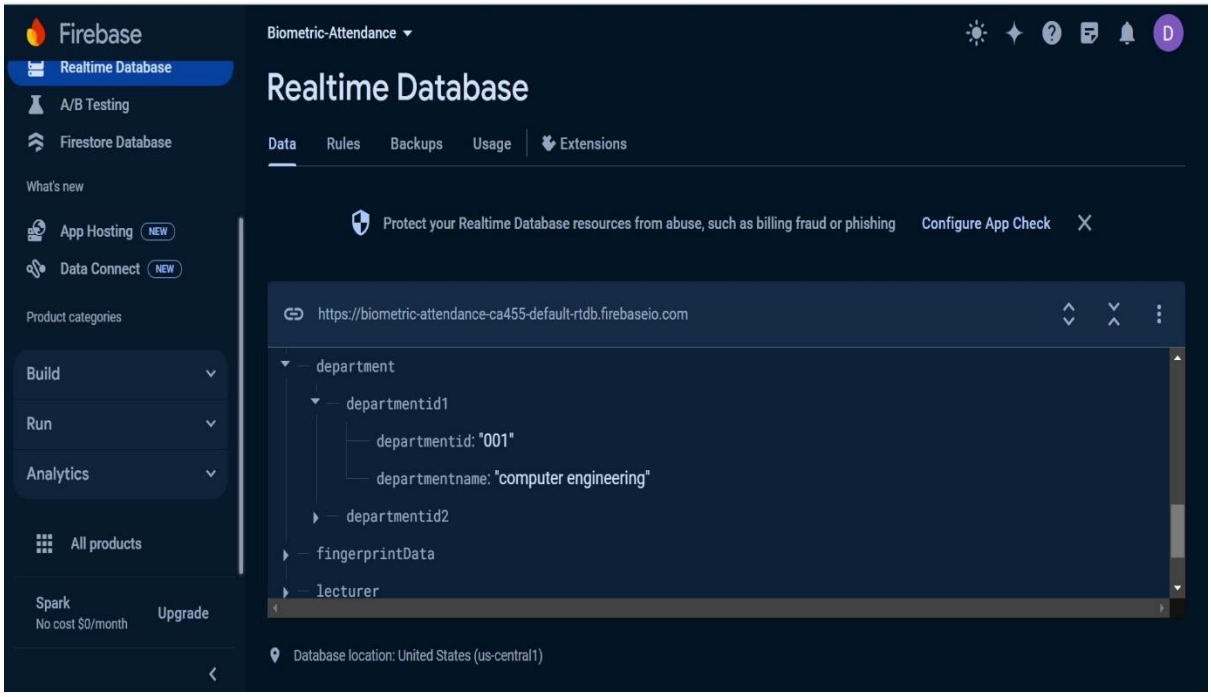
5. AttendanceRecord collection



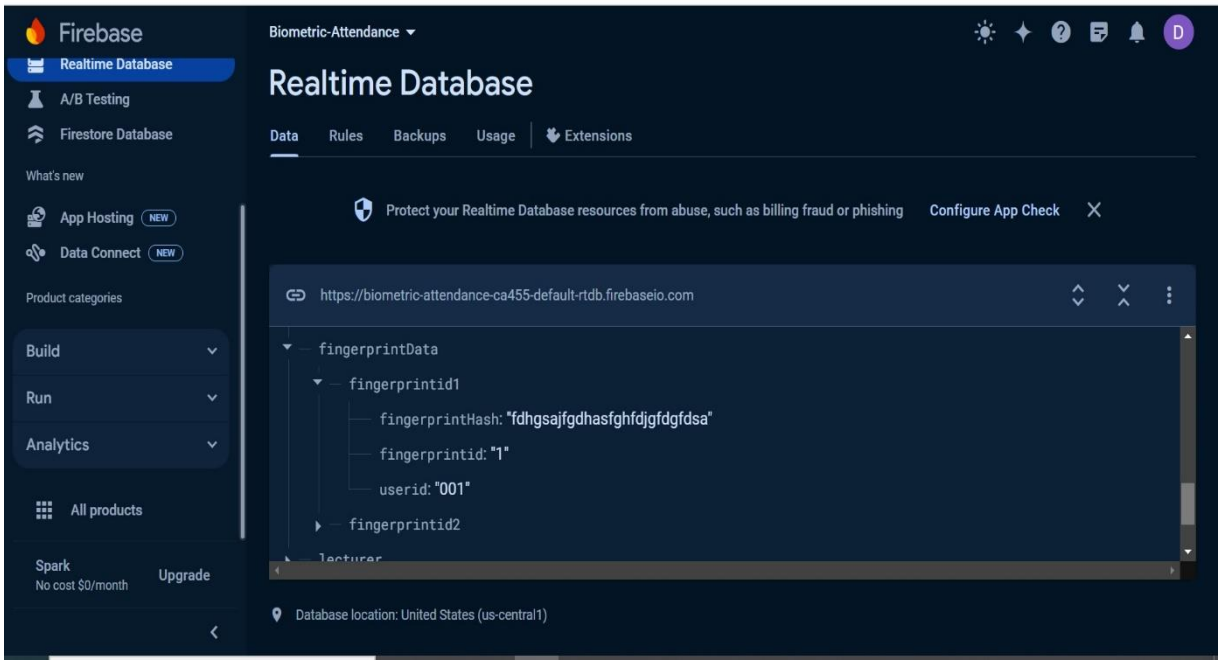
6. Course collection



7. Department collection



8. FingerprintData collection



4. CONCLUSION

This report presents a comprehensive database design for an attendance management system. By following a systematic design process, including requirement analysis, conceptual design, logical design, and normalization, the resulting database schema ensures efficient, secure, and organized management of student attendance. The detailed relational schema and ER diagram provide a clear roadmap for implementing the database. This design forms a robust foundation for developing a functional and reliable attendance management system.

5. REFERENCES

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- https://www.astera.com/type/blog/all-you-need-to-know-about-database-design/#:~:text=What%20is%20Database%20Design%3F,for%20the%20proposed%20database%20system.